Differences between the types of solutions.

Property	Suspension	Colloidal sol.	Solution
Particle size	>100nm	1 to100nm	<1nm
Filtration separation	Possible	Impossible	Impossible
Settling of particles	Settle on their own	Settle on centrifugation	Do not settle
Appearance	Opaque	Translucent (or) Semi transparent	Transparent
Scatterting of light	Does not penerate	Scatteres	Does not Scatter
Diffusion of particles	Do not diffuse	Diffuse slowly	Diffuse rapidly
Brownian movement	May show	Shows	Does not show
Nature	Heterogeneous	Heterogeneous	Homogeneous

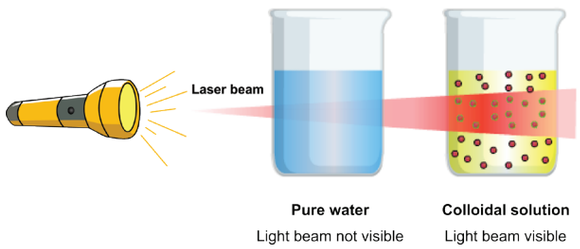


Figure 10.12 Tyndall effect

Test Yourself

1. Why whole milk is white?
2. Why ocean is blue?
3. Why sun looks yellow when it is really not?

10.4.3 Emulsions - a special kind of colloids

An emulsion is a colloid of two or more immiscible liquids where one liquid is dispersed in another liquid. This means one type of liquid particles get scattered in another liquid. In other words, an emulsion is a special type of mixture made by combining two liquids that normally don't mix. The word emulsion comes from the Latin word meaning "to milk" (milk is one example of an emulsion of fat and water). The process of turning a liquid mixture into an emulsion is called emulsification. Milk, butter, cream, egg yolk, paints, cough syrups, facial creams, pesticides etc. are some common examples of emulsions.

Types of emulsions

The two liquids mixed can form different types of emulsions. For example, oil and water can form an oil in water emulsion (O/W -e.g. cream), where the oil droplets are dispersed in water, or they can form a water in oil emulsion (W/O -e.g. butter), with water dispersed in oil.

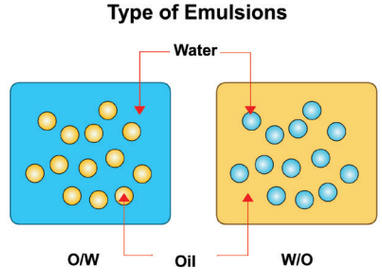


Figure 10.13 Emulsions

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